Lecture 28 – Inductors, LC, LR Circuits

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1 Intro to Inductors

When we include solenoids inside circuits, we call them 'inductors'. Their symbol takes the form of a line, wound into a loop. Inductance has the term 'L', measured with the Henry (H).

$$L := \frac{\phi_B}{I}$$

The B-field created by this is 'self-inducing', as it is created by the inductor itself.

1.1 Inductance of a simple solenoid

$$L = \frac{\phi_B}{I} = \frac{BA}{I}N$$

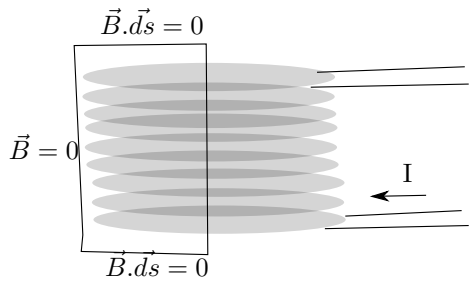

Figure 1: Simple solenoid

$$\oint \vec{B}.\vec{ds} = \mu_0 I_{thru}$$

$$Bl = \mu_0 NI$$

$$B = \frac{\mu_0 NI}{l}$$

$$L = \frac{\mu_0 N^2 IA}{lI}$$

$$L = \frac{\mu_0 N^2 A}{l}$$

1.2 Potential difference over an inductor

$$\Delta V = \oint \vec{E}.\vec{dS} = -\frac{d\phi_B}{dt}$$

But $\phi_B = LI$. Therefore,

$$\Delta V = -L\frac{dI}{dt}$$

This means a few things, which we will preface by recognizing that resistors 'resist' current by dropping V.

2 LR Circuits

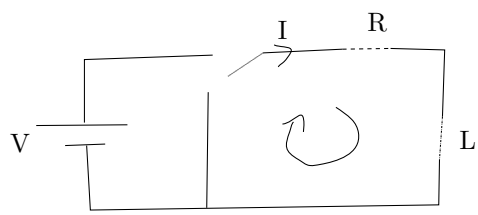

Figure 2: LR circuit 1

K2:

$$0 = -IR - L\frac{dI}{dt}$$

$$\frac{dI}{dt} = \frac{IR}{L}$$

$$\frac{dI}{dt}\frac{1}{I} = \frac{R}{L}$$

$$\int_0^t \frac{1}{I}\frac{dI}{dt}dt = \int_0^t -\frac{R}{L}dt$$

$$\int_{I(0)}^{I(t)} \frac{1}{I}dI =$$

$$[ln(I)]_{I_0}^{I_t} = [-\frac{R}{L}]_0^t$$

$$ln(I_t) - ln(I_0) = -\frac{R}{L}t$$

$$ln(\frac{I_t}{I_0}) = -\frac{R}{L}t$$

$$\frac{I_t}{I_0} = e^{-\frac{R}{L}t}$$

$$I(t) = I_0 e^{\frac{-t}{\tau}}$$

where $\tau = \frac{L}{R}$. So, as an overview:

LR Circuits

$I(t) = I_0 e^{\frac{-t}{\tau}}$, where $\tau = \frac{L}{R}$.

RC Circuits

$Q = Q_0 e^{\frac{-t}{\tau}}$, where $\tau = RC$.

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3 LC Circuits

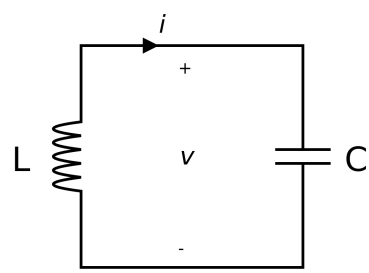

Figure 3: LC Circuit

The equation for energy in a capacitor is $\frac{1}{2}CV^2$, while that of an inductor is $\frac{1}{2}LI^2$. The total energy in the circuit is therefore:

$$E_{tot} = \frac{1}{2}CV^2 + \frac{1}{2}LI^2$$

Solving for Q/I

We know the equation for charge in a capacitor is $Q = CV$, and voltage across an inductor is Therefore, solving for K2:

$$0 = \frac{Q}{C} - L\frac{dI}{dt}$$

$$= \frac{Q}{C} + L\frac{d^2Q}{dt^2}$$

$$\frac{d^2Q}{dt^2} = -\frac{Q}{LC}$$

Therefore:

$$Q = Q_0 \cos \omega t$$

$$\omega = \sqrt{\frac{1}{LC}}$$